EV Battery Course — Lesson 1.1: What an Electric Vehicle Is and How It Works

Module 1 · Fundamentals of Electric Vehicles and Batteries · Oct 9, 2026 · @Ashishpreet

1. Learning objectives

By the end of this lesson you will be able to:

- Explain what an EV is in one sentence.
- Tell BEV, HEV, PHEV and FCEV apart.
- Trace how energy flows from the charger to the wheels.
- Name each major component and say what job it does.
- Explain why an EV turns stored energy into motion about three times more efficiently than an ICE car.

2. Fundamentals

An electric vehicle is driven, fully or partly, by an electric motor that takes its energy from an onboard store of electricity. In most EVs that store is a battery.

Compare it with a car you already know. In an ICE car, fuel is chemical energy: you burn it, use the heat to push a piston, and turn that linear motion into rotation. In an EV, the battery is also chemical energy, but it is released directly as electricity. The motor turns that electricity into rotation with no combustion step.

That one difference explains most of what makes an EV distinct:

- Efficiency is far higher, because nothing has to be converted to heat first.
- There are far fewer moving parts.
- Full torque is available from zero rpm.
- The hard engineering problem moves from the engine to the battery.

3. The four architectures

What separates the four types is what drives the wheels, where the energy comes from, and what job the battery does.

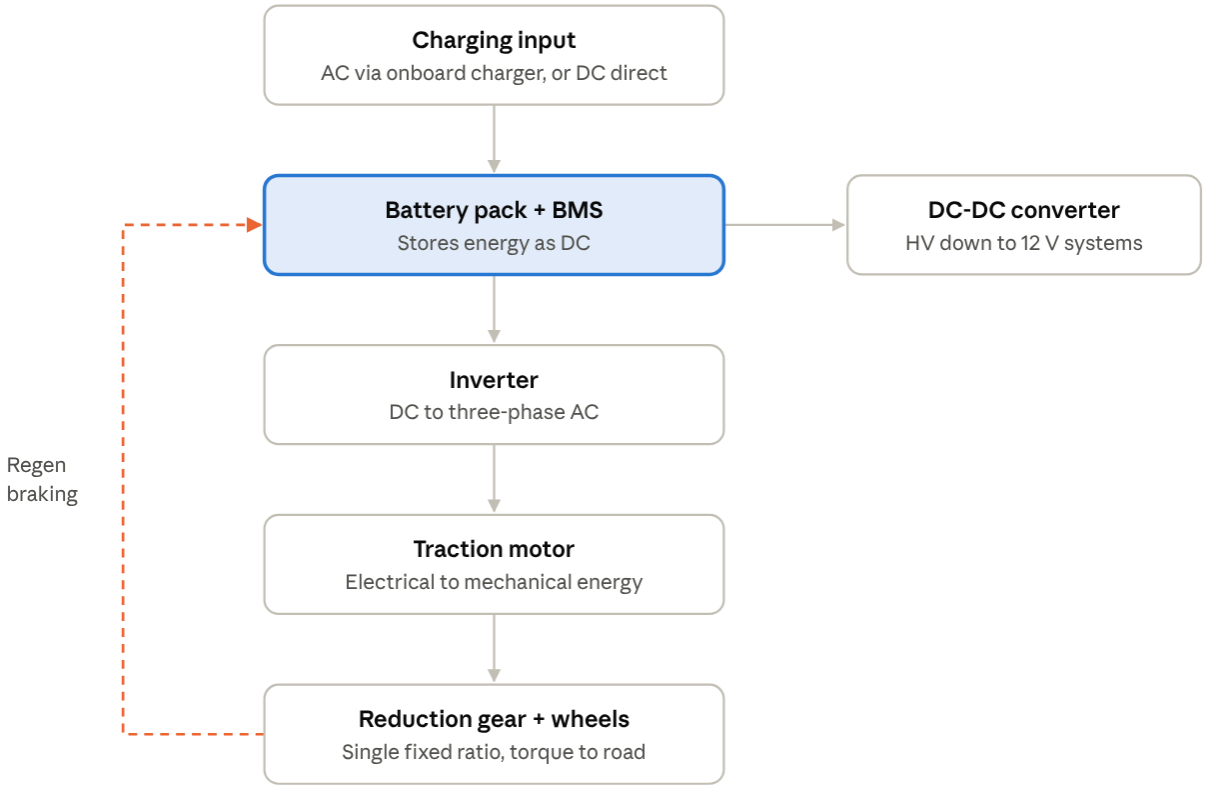
Type	What drives the wheels	Energy source	Plug in?	Battery role	Examples
BEV (battery electric)	Motor only	Battery only	Yes	Energy store, large (30–100+ kWh)	Tata Nexon EV, MG ZS EV, Ola S1
HEV (strong hybrid)	Engine + motor	Petrol only; battery recharged by engine and braking	No	Power buffer, small (about 1–2 kWh)	Toyota Innova Hycross, Maruti Grand Vitara strong hybrid
PHEV (plug-in hybrid)	Engine + motor	Grid and petrol	Yes	Both roles, medium (about 10–25 kWh, 40–80 km electric range)	Mitsubishi Outlander PHEV, BYD DM-i models
FCEV (fuel-cell electric)	Motor only	Hydrogen turned into electricity onboard	Hydrogen refuel	Small buffer battery	Toyota Mirai, Hyundai Nexo

The key distinction is between a power battery and an energy battery. An HEV battery is cycled shallowly thousands of times a day and needs high power for its size. A BEV battery needs a lot of stored energy. Later lessons show how these two needs lead to different cell designs.

A mild hybrid (48 V start-stop with a small assist motor) is not an EV in any meaningful sense, because the motor cannot drive the car on its own.

4. How energy flows in a BEV

Energy flows from the battery to the wheels; regen sends it back



BEV energy flow · charger to wheels, with regen

Energy enters as AC or DC, is stored as DC in the pack, and becomes three-phase AC only at the inverter. Under braking the motor runs as a generator and pushes energy back into the battery.

5. Step-by-step working principle (BEV)

1. **Charging.** Mains AC goes through the onboard charger (OBC), which converts it to DC to charge the battery. A DC fast charger does that conversion outside the car and feeds DC straight to the pack, which is why it can deliver much more power.
2. **Storage.** The battery stores the energy chemically: lithium ions held in the anode represent stored energy. The battery management system (BMS) monitors every cell's voltage and temperature and decides how much current the pack may safely supply or accept.
3. **Pressing the accelerator.** The vehicle controller asks the inverter for torque. The inverter switches battery DC thousands of times a second to build three-phase AC at the frequency and amplitude the motor needs.

4. **Motor.** The rotating magnetic field in the stator pulls the rotor around, producing torque from 0 rpm.
5. **Reduction gear.** Motors spin at up to about 15,000–20,000 rpm, while wheels turn at roughly 1,000–1,500 rpm at highway speed. A single fixed ratio of about 8–10:1 is enough, so there is no clutch and no multi-speed gearbox.
6. **Regenerative braking.** When you lift off the accelerator, the motor works as a generator. The inverter rectifies its output back to DC and recharges the battery.
7. **Auxiliaries.** The DC-DC converter steps the high-voltage pack (about 300–800 V) down to 12 V. That powers the lights, ECUs and infotainment, and keeps the small 12 V battery charged.

6. Engineering equations

Range

$$\text{Range (km)} = \frac{E_{usable}\ (\text{Wh})}{\text{Consumption}\ (\text{Wh/km})}$$

E_usable is the energy the BMS lets you use, always less than the gross energy. Consumption is typically 100–180 Wh/km for passenger cars and 25–40 Wh/km for two-wheelers.

Efficiency of the drive chain

$$\eta_{drive} = \eta_{inverter} \times \eta_{motor} \times \eta_{gear}, \qquad P_{wheel} = P_{battery} \times \eta_{drive}$$

η is efficiency as a fraction and P is power in kW. When components are in series you multiply their efficiencies, exactly as you would for a gearbox and bearing train.

Kinetic energy (the energy available to regen)

$$KE = \tfrac{1}{2} m v^2$$

m is mass in kg, v is speed in m/s, and KE is in joules. 1 kWh = 3.6 × 10^6 J.

7. Worked examples

Example 1: range. A compact SUV has a 45 kWh gross pack. The BMS allows about 93% to be used, and real-world consumption is 150 Wh/km.

- E_usable = 45,000 × 0.93 = 41,850 Wh
- Range = 41,850 ÷ 150 ≈ **279 km**

Example 2: drive-chain losses. Inverter efficiency is 97%, motor 93% and gear 97%, with the battery delivering 20 kW while cruising.

- $\eta = 0.97 \times 0.93 \times 0.97 =$ **0.875 (87.5%)**
- Power at the wheels = $20 \times 0.875 =$ **17.5 kW**. The other 2.5 kW becomes heat, mostly in the motor, and the cooling system has to remove it.

Example 3: why EVs are efficient. Petrol holds about 8.9 kWh of chemical energy per litre, and a petrol engine plus transmission delivers roughly 25% of that to the wheels in real driving.

- 45 kWh of battery energy equals only $45 \div 8.9 \approx$ **5 L of petrol**.
- It delivers $45 \times 0.875 \approx 39$ kWh at the wheels. An ICE car would need $39 \div (8.9 \times 0.25) \approx$ **17.5 L** to do the same.

The battery's real weakness is its low energy per kilogram, not how efficiently that energy is used. That is why most of this course is about the battery.

8. Real-world examples

- **Passenger BEV:** The Tata Nexon EV is a typical Indian layout: a front motor, a single reduction gear, an LFP pack under the floor and a liquid-cooled thermal system.
- **Two-wheeler:** Ola S1 and Ather scooters use packs of about 3–4 kWh. The low mass means consumption of roughly a quarter of a car's per km.
- **Strong hybrid:** The Toyota/Maruti hybrid system (Hycross, Hyryder, Grand Vitara) runs the engine near its best-efficiency point and uses a small battery to fill the gaps. Customers never plug in.
- **Bus:** Olectra and Tata electric buses carry packs of about 250–350 kWh. Depot charging and predictable routes suit BEVs well.
- **FCEV:** India trialled a Toyota Mirai in 2022, but hydrogen refuelling infrastructure is close to non-existent today.

9. Manufacturing implications

- **Fewer parts, different parts.** There is no engine block machining, crankshaft, fuel system, exhaust or multi-speed gearbox. Instead the plant builds or buys the battery pack, inverter, motor, high-voltage harness and thermal plates.
- **Cost concentrates.** The battery pack is the largest single BOM item, commonly cited at around 30–40% of a mass-market BEV's cost, and falling as cell prices drop. For should-cost work, one commodity (the cells) drives most of the variance.

- **New process controls.** The plant needs HV insulation-resistance tests, high-voltage interlock checks, end-of-line testing of BMS functions, HV-safe tooling and trained technicians. Cell storage also falls under dangerous-goods handling.
- **Weight shifts to the floor.** A 300–450 kg pack changes body-in-white design, crash load paths, and the load ratings of paint-shop and assembly conveyors.

10. Common misconceptions

- **"A hybrid is a kind of EV you charge at home."** Only a PHEV can be plugged in. An HEV never is.
- **"The battery outputs AC to the motor."** Batteries are always DC. The inverter creates the AC.
- **"A bigger battery always means more range."** The extra mass raises consumption, so the gain is less than proportional.
- **"EVs don't have a 12 V battery."** They do. In many EVs a flat 12 V battery stops the car from waking up even when the main pack is full.
- **"Regen recovers all braking energy."** In practice it recovers perhaps 50–70% of the kinetic energy available during a braking event. The rest is lost to friction brakes, the motor, the inverter and the battery.

11. Case study: "my car doesn't reach its rated range"

Problem. Owners report about 220 km of real range against a homologated figure of about 300+ km, and warranty and complaint costs rise.

Root causes.

1. The Indian test cycle (MIDC) is gentle, with low average speeds.
2. Aerodynamic drag rises with v^2, so driving at 100 km/h costs far more per km than city driving.
3. AC compressor load in Indian summers.
4. Battery and motor heating, which adds cooling load.

Engineering response. Publish a real-world range alongside the certified figure. Add eco and regen-level modes. Use heat pumps and better thermal management. Improve the range-estimate algorithm so it learns from the driver's habits.

Lesson. Range is a vehicle-system outcome, not just a battery specification.

12. Revision notes and glossary

Revision notes

- An EV turns battery DC into AC through the inverter, then into rotation in the motor, then delivers it through a single-speed gear.
- BEV = battery only. HEV = engine with a small power battery and no plug. PHEV = both, with a plug. FCEV = hydrogen made into electricity onboard.
- A BEV is about 85–90% efficient from battery to wheels, against about 20–30% for an ICE from tank to wheels.
- The battery stores little energy per kg, so it dominates cost, mass and engineering effort.
- Range = usable energy ÷ consumption.

Glossary

Term	Meaning
OBC	Onboard charger; converts AC to DC
BMS	Battery management system; the battery's monitor and protector
Inverter	Converts DC to variable-frequency AC for the motor
DC-DC converter	Steps high voltage down to 12 V
Regen	Regenerative braking; the motor acts as a generator
HV	High voltage; in automotive terms, above 60 V DC
MIDC	Modified Indian Driving Cycle, used for homologation
Gross vs usable energy	Total stored energy vs what the BMS allows you to draw

13. Knowledge check

Try these before turning to the answer key.

Conceptual

1. Why does an HEV not need a charging port?

2. How does the battery's job differ between an HEV and a BEV?
3. Why does an EV need an inverter if the battery is DC?
4. Why does a BEV still carry a 12 V battery?
5. Why can most EVs use a single-speed gear?

Numerical

1. A car has 30 kWh usable and uses 125 Wh/km. What is its range?
2. The inverter is 96% efficient, the motor 94% and the gear 98%. The battery supplies 50 kW. What reaches the wheels, and how much is lost?
3. A 1,500 kg car brakes from 72 km/h to rest, and 60% of its kinetic energy is recovered. How much energy is that in Wh, and how many km does it add at 150 Wh/km?

Application

1. You are building a should-cost model for a BEV versus its ICE twin. Which three systems would you model in most detail, and why?
2. A Gurgaon fleet runs 200 km a day in city traffic and parks overnight at a depot. Which architecture would you recommend?

14. Answer key

Conceptual

1. All of its energy comes from petrol. The engine and regen braking recharge the small battery.
2. An HEV battery is a power buffer: small, cycled shallowly and very often. A BEV battery is the energy tank: large and deeply cycled.
3. Efficient traction motors (PMSM and induction) need variable-frequency AC to control speed and torque. The inverter synthesises that AC from DC.
4. ECUs, lights, locks and the contactors that connect the HV pack all run on 12 V. The car has to boot before the HV pack is switched in.
5. An electric motor delivers usable torque across a very wide speed range, from 0 to about 15,000+ rpm, so one ratio covers the whole vehicle speed range.

Numerical

1. 30,000 ÷ 125 = **240 km**.
2. $\eta = 0.96 \times 0.94 \times 0.98 = 0.884$. Power at the wheels = 50 × 0.884 = **44.2 kW**. Losses = **5.8 kW**, which the coolant has to carry away.
3. $v = 72 \div 3.6 = 20$ m/s. $KE = \frac{1}{2} \times 1{,}500 \times 20^2 = 300{,}000$ J = 0.0833 kWh. Recovered = 0.6 × 83.3 ≈ **50 Wh**, which gives 50 ÷ 150 ≈ **0.33 km**. One stop recovers little, but city traffic with hundreds of stops a day adds up to a meaningful share of range.

Application

1. Model these three in most detail:
   - The battery pack: the largest cost line, driven by cell price per kWh, pack integration and the thermal system.
   - The e-drive (motor, inverter and gear): magnets, copper and power semiconductors.
   - The HV system (harness, OBC, DC-DC and junction box): new commodities with no ICE equivalent.

   On the ICE side, the costs that disappear are the engine, transmission, fuel system and exhaust after-treatment.
2. A BEV. The duty cycle is predictable, overnight depot charging is available, and stop-start city driving gives regen plenty to recover. Daily energy need is roughly 200 × 0.15 = 30 kWh, comfortably within a mid-size pack, and operating cost per km is much lower than petrol or CNG. A strong hybrid is the fallback only if depot charging isn't possible.

15. Further learning

- *Electric Vehicle Technology Explained*, Larminie & Lowry (Wiley): the best first textbook for this module.
- *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*, Ehsani et al.: deeper powertrain engineering.
- NPTEL, "Introduction to Hybrid and Electric Vehicles" (IIT Guwahati): free and India-relevant.
- Battery University (batteryuniversity.com): short, readable articles on chemistry and cell behaviour.
- Indian standards: AIS-038 Rev 2 (EV electric powertrain safety, cars) and AIS-156 (two- and three-wheelers). Both return in the safety and testing modules.

Next: Lesson 1.2 covers voltage, current, resistance, power vs energy, Ah vs kWh, and series vs parallel.